

# Nexlara Deal Health Copilot — Full Build Specification

**Version:** 2.0 (Updated)  
**Last Updated:** June 24, 2026  
**Platform:** Oracle ERP Cloud + Oracle CPQ

## 1. Project Overview

**Product:** Nexlara Intelligent Configuration Copilot

**Purpose:** A web application prototype demonstrating an AI-powered, real-time quote configuration intelligence layer for Nexlara Technologies' deal approval process. It embeds an 8-rule validation engine directly into the Oracle CPQ configuration experience so AEs get live guidance DURING configuration, not after submission.

**Platform:** Oracle ERP Cloud + Oracle CPQ (migrated from D365 — full API exposure via Oracle REST endpoints)

**Stack:** React 19 + TypeScript + Tailwind CSS 4 + shadcn/ui + Wouter (routing) + Recharts + Framer Motion + Lucide Icons + Sonner (toasts)

**Fonts:** Space Grotesk (display/headlines), Inter (body), JetBrains Mono (monospace data/IDs)

## 2. Design System — "Mission Control"

### Aesthetic

Aerospace-inspired dark command center. Bloomberg Terminal density meets Linear/Vercel dark-mode SaaS. Operational urgency, not decorative.

### Color System (OKLCH-based CSS variables)

| Token             | Value                 | Hex Approx | Usage                           |
|-------------------|-----------------------|------------|---------------------------------|
| Canvas/Background | oklch(0.13 0.02 280)  | #0F1117    | Page background                 |
| Surface/Card      | oklch(0.16 0.025 275) | #1A1D2E    | Elevated panels                 |
| Primary Accent    | —                     | #A100FF    | Nexlara brand (Electric Violet) |

|                |                      |         |                                                                 |
|----------------|----------------------|---------|-----------------------------------------------------------------|
| Success        | —                    | #00E676 | Auto-approved, healthy                                          |
| Warning        | —                    | #FFB300 | Needs attention                                                 |
| Danger         | —                    | #FF5252 | Rejected, critical, DHI blocker                                 |
| Text Primary   | oklch(0.92 0.01 250) | #E8EAED | Main content                                                    |
| Text Secondary | oklch(0.65 0.02 270) | #9CA3AF | Muted/labels                                                    |
| Borders        | oklch(0.25 0.04 280) | —       | Purple-tinted with <code>rgba(161, 0, 255, 0.15)</code> accents |

## Signature Visual Elements

1. **Glowing Score Rings** — circular progress with gradient strokes
2. **Status Beacon Dots** — small colored dots as ambient health indicators
3. **Rule Activity Bar** — shows which of 8 validation rules just fired
4. **Purple edge glow on panels:** `box-shadow: 0 0 20px rgba(161, 0, 255, 0.15)`

## Typography Hierarchy

| Level     | Font           | Size    | Weight                    |
|-----------|----------------|---------|---------------------------|
| H1        | Space Grotesk  | 18–20px | Bold                      |
| H2        | Space Grotesk  | 14–16px | Semibold                  |
| Body      | Inter          | 13–14px | Regular                   |
| Monospace | JetBrains Mono | 10–12px | Regular                   |
| Captions  | Inter          | 9–11px  | Uppercase, tracking-wider |

## Layout Paradigm

- Persistent collapsible left sidebar (68px collapsed / 240px expanded)
- Main content uses asymmetric panels — wide workspace + narrower intelligence panel on right

- Cards use `border border-border/50` with subtle purple glow on hover

### 3. Application Architecture

#### Two Modules, Same App

| Module          | Base Route             | Primary Persona                | Purpose                                                          |
|-----------------|------------------------|--------------------------------|------------------------------------------------------------------|
| AE Module       | <code>/</code>         | Account Executive (Sarah Chen) | Quote configuration, deal health monitoring, pipeline management |
| Customer Portal | <code>/customer</code> | Enterprise Customer            | Proposal submission, status tracking, history                    |

#### Routing Structure

| Plain Text                          |                                              |
|-------------------------------------|----------------------------------------------|
| <code>/</code>                      | → AE Dashboard (command center)              |
| <code>/quote-builder</code>         | → AE Quote Configuration (PRIMARY DEMO PAGE) |
| <code>/deal/:id</code>              | → AE Deal Detail                             |
| <code>/pipeline</code>              | → AE Pipeline View                           |
| <code>/compliance</code>            | → AE Compliance View                         |
| <code>/analytics</code>             | → AE Analytics                               |
| <code>/customer</code>              | → Customer My Proposals                      |
| <code>/customer/submit</code>       | → Customer New Proposal                      |
| <code>/customer/proposal/:id</code> | → Customer Proposal Detail                   |
| <code>/customer/history</code>      | → Customer History                           |

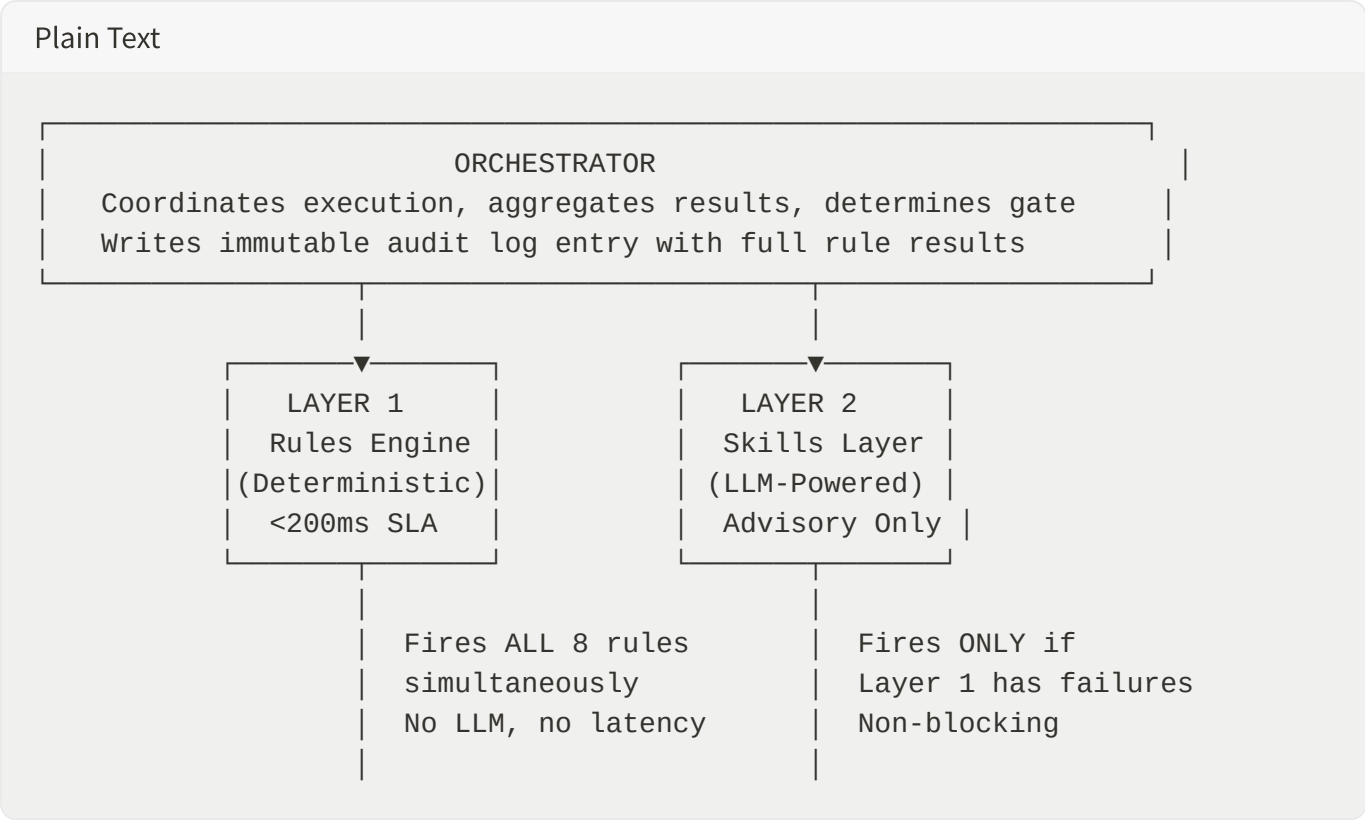
#### Module Switching

- AE sidebar bottom: "Switch to Customer Portal" link → `/customer`
- Customer sidebar bottom: "Switch to AE View" link → `/`

### 4. AI Agent Architecture — 2-Layer Orchestrated System

#### Overview

The Nexlara Copilot uses a **deterministic-first, LLM-second** architecture. The Rules Engine (Layer 1) makes all binding pass/block decisions. The Skills Layer (Layer 2) provides human-readable explanations only when rules fail.



### 4.1 Layer 1 — Rules Engine (8 Deterministic Rules)

**CRITICAL DESIGN PRINCIPLE:** All validation fires DURING configuration, not at submission time. The intelligence is embedded in the configuration experience itself. All rules execute simultaneously with a <200ms latency target. No LLM involvement — pure deterministic logic.

| # | Rule ID   | Name               | Severity | Trigger                       | What It Validates                                                   |
|---|-----------|--------------------|----------|-------------------------------|---------------------------------------------------------------------|
| 1 | LEP_001   | LEP Compliance     | BLOCKER  | SKU added or discount changes | All BOM line prices are at or above the Lowest Entitled Price floor |
| 2 | PHASE_001 | Year-1 ACV Phasing | BLOCKER  | Year-1 ramp slider adjusted   | Year-1 revenue is ≥50% of total contract value                      |

|   |                |                              |                |                          |                                                                       |
|---|----------------|------------------------------|----------------|--------------------------|-----------------------------------------------------------------------|
| 3 | ASC606_001     | ASC 606 SSP Validation       | <b>BLOCKER</b> | Non-perpetual SKU added  | Standalone Selling Price documented for all consumption/overage lines |
| 4 | MCCA_001       | MCCA Status Check            | <b>BLOCKER</b> | Any configuration change | Master Cloud Customer Agreement is signed before submission           |
| 5 | DHI_001        | Deal Health Index (DHI)      | <b>BLOCKER</b> | Any configuration change | Aggregates all critical/high flags — blocks if any unresolved         |
| 6 | AUTOATTACH_001 | Auto-Attach AI SKU Detection | WARNING        | SKU added to BOM         | Detects if AI add-on SKUs should be attached to base products         |
| 7 | DEPRECATED_001 | Deprecated SKU Check         | WARNING        | SKU added to BOM         | Flags any end-of-life SKUs in the BOM                                 |
| 8 | DOA_001        | DoA Gate Threshold           | <b>BLOCKER</b> | Any configuration change | Determines approval routing tier based on deal value + risk           |

## Rule Severity Definitions

| Severity       | Behavior                                                                | Count   |
|----------------|-------------------------------------------------------------------------|---------|
| <b>BLOCKER</b> | Prevents quote submission entirely. Must be resolved before proceeding. | 5 rules |
| <b>WARNING</b> | Advisory flag shown to AE. Does not block submission.                   | 2 rules |

|      |                                                                     |        |
|------|---------------------------------------------------------------------|--------|
| GATE | Routing determination — always passes but determines approval path. | 1 rule |
|------|---------------------------------------------------------------------|--------|

## Gate Decision Logic

After all 8 rules complete:

- If **0 BLOCKERS failed** → GATE: CLEAN → proceeds to DoA routing
- If **any BLOCKER failed** → GATE: BLOCKED → submission halted, AE must resolve

## DoA Routing Tiers (Rule DOA\_001)

| Tier           | Name        | Conditions                                        | Action                                |
|----------------|-------------|---------------------------------------------------|---------------------------------------|
| AUTO_APPROVE   | Instant     | Deal <€500K, 0 flags, discount ≤15%, ML score ≥90 | System auto-approves, no human review |
| FIRST_APPROVER | Standard    | Deal <€500K with flags                            | Routes to first-line approver         |
| SUPER_APPROVER | Elevated    | Deal ≥€500K                                       | Routes to senior approver             |
| MU_CFO         | Executive   | Deal ≥€2M                                         | Routes to Market Unit CFO             |
| BOARD          | Board-Level | Deal ≥€5M                                         | Routes to Board review                |

## 4.2 Layer 2 — Skills Layer (LLM-Powered, Advisory Only)

The Skills Layer fires **only when Layer 1 has failures or warnings**. It does NOT make binding decisions — it translates rule outputs into human-readable guidance.

| Skill             | Trigger Condition          | Purpose                                                            | Output                                            |
|-------------------|----------------------------|--------------------------------------------------------------------|---------------------------------------------------|
| DHI Translator    | Any BLOCKER fails          | Converts cryptic rule codes into plain-English guidance for the AE | Natural language explanation + specific fix steps |
| Narrative Advisor | Justification weak/missing | Suggests business justification language                           | Recommended justification text                    |

|              |                      |                                                                         |                                            |
|--------------|----------------------|-------------------------------------------------------------------------|--------------------------------------------|
|              |                      | to strengthen the deal case                                             |                                            |
| ML Explainer | ML score < threshold | Explains why the ML model scored the deal low and what would improve it | Factor breakdown + improvement suggestions |

## Skills Layer Constraints

- **Non-blocking:** Skills never prevent submission; they only advise
- **Latency budget:** 2–5 seconds (acceptable because advisory, not gating)
- **Fallback:** If LLM is unavailable, raw rule output is shown instead
- **No hallucination risk:** Skills only interpret existing rule results, never generate new validation logic

## 4.3 Orchestrator Behavior

The Orchestrator coordinates the full execution flow when "Submit to Approval" is triggered:

1. **Initialize** — Display processing modal, set all rules to "pending"
2. **Layer 1 Execution** — Fire all 8 rules simultaneously (<200ms)
3. **Gate Determination** — Count BLOCKER failures → set CLEAN or BLOCKED
4. **Layer 2 Execution** (conditional) — If any failures/warnings exist, fire relevant Skills
5. **DoA Routing** (if CLEAN) — Determine approval tier from DOA\_001 result
6. **Audit Entry** — Write immutable log with: timestamp, all rule results, gate status, DoA tier, actor, quote ID
7. **Result Display** — Show final summary with pass/fail per rule, gate status, and next steps

## Orchestrator Output Schema

TypeScript

```
interface OrchestratorResult {
  quoteId: string;
  timestamp: string;
  gateStatus: "CLEAN" | "BLOCKED";
  doaTier: "AUTO_APPROVE" | "FIRST_APPROVER" | "SUPER_APPROVER" | "MU_CFO" | "B
  rules: {
```

```

    id: string;
    name: string;
    severity: "BLOCKER" | "WARNING";
    status: "pass" | "fail" | "warning";
    message: string;
    executionMs: number;
  }[];
  skills: {
    name: string;
    triggered: boolean;
    output: string;
  }[];
  auditId: string;
  actor: string;
}

```

## 4.4 Test Scenarios (from documented test\_runner)

| Test Case     | Quote ID     | Expected Gate | Expected DoA | Key Condition                            |
|---------------|--------------|---------------|--------------|------------------------------------------|
| Clean Pass    | Q-TEST-CLEAN | CLEAN         | AUTO_APPROVE | All rules pass, <€500K, 0 flags          |
| LEP Breach    | Q-2025-00001 | BLOCKED       | —            | Discount 22% breaches LEP floor          |
| Multi-Failure | Q-2025-00005 | BLOCKED       | —            | LEP + MCCA + DHI all fail simultaneously |

## 5. Platform Integration — Oracle ERP Cloud + Oracle CPQ

### 5.1 System of Record

**Oracle CPQ** is the system of record for all quote configuration. The Nexlara Copilot is an **extension layer** — it does not replace Oracle CPQ, it enhances the configuration experience with real-time intelligence.

### 5.2 Integration Pattern

| Integration Point | Technology | Purpose |
|-------------------|------------|---------|
|-------------------|------------|---------|

|                     |                                                      |                                                 |
|---------------------|------------------------------------------------------|-------------------------------------------------|
| Quote Events        | Oracle Business Events + OIC                         | Trigger rules on configuration changes          |
| Price Validation    | Oracle CPQ <code>calculatePrice</code> REST API      | Validate LEP compliance in real-time            |
| BOM Validation      | Oracle CPQ <code>validateBomItemTree</code> endpoint | Check SKU dependencies and deprecated items     |
| Customer Data       | Oracle ERP REST API (Accounts/Contacts)              | Retrieve MCCA status, account history           |
| Revenue Recognition | Oracle Revenue Management Cloud                      | ASC 606 SSP validation                          |
| Approval Routing    | Oracle Integration Cloud (OIC) flows                 | DoA tier routing and notification               |
| Audit Logging       | Oracle Autonomous Database (immutable)               | Full audit trail with actor/timestamp/rationale |

## 5.3 Oracle CPQ BML Integration

For real-time (synchronous) validation during configuration, Business Management Language (BML) scripts execute within Oracle CPQ:

Plain Text

```
// Pseudocode – LEP Compliance Rule (BML)
if (lineItem.unitPrice < lineItem.lepFloor) {
  addValidationError("LEP_001", "BLOCKER",
    "Unit price " + lineItem.unitPrice + " is below LEP floor " + lineItem.lepF
}
```

## 5.4 Key Oracle API Endpoints Used

| Endpoint                                                                                  | Method   | Purpose                     |
|-------------------------------------------------------------------------------------------|----------|-----------------------------|
| <code>/rest/v17/commerceDocumentsOraclecpqoTransaction</code>                             | GET/POST | Quote CRUD operations       |
| <code>/rest/v17/commerceDocumentsOraclecpqoTransaction/{id}/actions/calculatePrice</code> | POST     | Real-time price calculation |

|                                                                                   |      |                               |
|-----------------------------------------------------------------------------------|------|-------------------------------|
| /rest/v17/commerceDocumentsOraclecpqoTransaction/{id}/actions/validateBomItemTree | POST | BOM validation                |
| /fscmRestApi/resources/latest/receivablesInvoices                                 | GET  | Revenue recognition data      |
| /ic/api/integration/v1/flows                                                      | POST | Trigger OIC integration flows |

## 6. The 8 Non-Negotiable Real-Time Validation Rules (Detailed)

### Rule 1: LEP Compliance (LEP\_001) — BLOCKER

**Trigger:** The moment each SKU is added or discount percentage changes.

**Logic:** For each BOM line item, compare  $\text{unitPrice} \times (1 - \text{discount}\%)$  against the LEP floor price from the SKU catalogue. If any line falls below LEP, the rule fails.

**Display:** Live price-vs-LEP check per line item with red/green indicator.

**Auto-Fix:** If discount causes breach, suggest maximum allowable discount percentage.

### Rule 2: Year-1 ACV Phasing (PHASE\_001) — BLOCKER

**Trigger:** As the seller adjusts the Year-1 ramp slider.

**Logic:**  $\text{Year1ACV} / \text{TotalContractValue} \geq 0.50$  . Must be at least 50%.

**Display:** Year-1 ACV % displayed continuously with threshold marker at 50%.

### Rule 3: ASC 606 SSP Validation (ASC606\_001) — BLOCKER

**Trigger:** The moment a non-perpetual SKU is added to the BOM.

**Logic:** Verify Standalone Selling Price documentation exists for all consumption/overage lines. Check performance obligation classification. Validate variable consideration estimates.

**Display:** SSP documentation status, performance obligation count, variable consideration warnings.

### Rule 4: MCCA Status Check (MCCA\_001) — BLOCKER

**Trigger:** Any configuration change (checked continuously).

**Logic:** Query Oracle ERP customer record for MCCA agreement status. Must be "Signed" or "Active" to proceed.

**Display:** MCCA status badge (Signed/Pending/Expired) with link to agreement record.

## Rule 5: Deal Health Index — DHI (DHI\_001) — BLOCKER

**Trigger:** Any configuration change (aggregates all other rule outputs).

**Logic:** Counts all unresolved critical and high-severity flags across all rules. If `unresolvedCriticalFlags > 0`, the DHI rule fails and blocks submission.

**Display:** Aggregate health score (0–100) with breakdown of contributing factors. Plain-language flags with specific resolution guidance and "Resolve Inline" buttons.

**Non-Negotiable Constraint:** DHI is a hard blocker. No quote can proceed to approval with unresolved critical flags regardless of deal value or executive override.

## Rule 6: Auto-Attach AI SKU Detection (AUTOATTACH\_001) — WARNING

**Trigger:** When a base product SKU is added that has associated AI add-on SKUs.

**Logic:** Check if the added product has recommended AI add-on SKUs that should be attached. Flag for AE awareness but do not block.

**Display:** Advisory banner suggesting AI SKU attachment with "Accept" or "Dismiss" options.

## Rule 7: Deprecated SKU Check (DEPRECATED\_001) — WARNING

**Trigger:** When any SKU is added to the BOM.

**Logic:** Cross-reference SKU against the catalogue's `status` field. Flag any SKU with status "Deprecated" or "End-of-Life" and suggest successor SKU.

**Display:** Warning badge on deprecated line items with successor SKU recommendation and "Replace" button.

## Rule 8: DoA Gate Threshold (DOA\_001) — BLOCKER

**Trigger:** Any configuration change (recalculates continuously).

**Logic:** Based on total deal value, discount percentage, flag count, and ML approval score, determine which DoA tier the deal routes to. The tier itself is informational, but the rule ensures the routing is always calculated and displayed.

**Display:** DoA tier badge with threshold explanation and approver identity.

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## 7. Page Specifications

## 7.1 AE Dashboard (/)

**Layout:** Dense command center with horizontal metrics strip + active deals workspace + intelligence panel.

### Components:

- **Metrics Strip** (horizontal row): Cycle Time, Auto-Approval Rate, First-Pass Rate, Cost/Quote, QE Deals pending — each with trend indicator
- **Active Deals Table:** Sortable list showing deal name, customer, value, health score (circular ring), stage badge, days in stage, flag count
- **Copilot Intelligence Panel** (right side): Risk alerts, contextual recommendations, recent activity feed
- **HUD elements:** Faint grid lines, beacon dots, coordinate-style labels

## 7.2 AE Quote Builder (/quote-builder) — PRIMARY DEMO PAGE

**Layout:** Full-width form with intelligence panel below.

### Header Bar:

- User context: Sarah Chen • AE • EMEA-DACH
- System indicator: ● 8 RULES ACTIVE
- Platform badge: Oracle CPQ

### LIVE RULES Bar:

Horizontal strip showing all 8 rules with colored beacon dots. When a rule fires, it flashes/highlights for 2 seconds to show which rule just reacted to the user's action.

Plain Text

LIVE RULES: ⚡ LEP Compliance | 📊 Phasing Validator | 📋 ASC 606 SSP | 🔒 MCCA

### Configuration Form:

- Customer & Opportunity fields (pre-filled, linked to Oracle Opportunity)
- License Model selector (Subscription / Consumption / Hybrid / Perpetual)
- Contract Term selector (12/24/36/48/60 months)
- Discount % input with live LEP check
- Year-1 ACV % slider (phasing) with live validation and 50% minimum marker
- BOM Editor: Add/remove SKU lines with product dropdown (66-item Nexlara catalogue), quantity, unit price, LEP indicator

- Business Justification textarea
- Competitive Context textarea
- Action buttons: Save Draft / Submit to Approval

#### Intelligence Panel (below form):

- **Approval Probability Ring:** Large circular score with color (green  $\geq 85$ , amber  $\geq 65$ , red  $< 65$ )
- **Top Driving Factors:** 2–3 items showing what's pushing score up/down with impact values
- **DoA Routing:** Which approval tier the deal routes to, with threshold explanation
- **Auto-Approval Eligibility:** Green badge if score  $\geq 90$  and no critical/high flags
- **DHI Flags:** Each flag shows code, severity color, plain-language title, description, fix guidance, and "Resolve Inline" button if auto-fixable
- **ASC 606 Checks:** Rule name, pass/fail/warning status, detail, and what triggered it

#### On Submit — AI Agents Processing Modal:

Full-screen modal showing the 2-layer orchestrator executing:

1. Layer 1: All 8 rules fire simultaneously with animated progress bars
2. Gate determination: CLEAN or BLOCKED displayed prominently
3. Layer 2: Skills fire conditionally (DHI Translator, Narrative Advisor, ML Explainer)
4. Final summary: DoA tier, audit ID, next steps

### 7.3 AE Deal Detail (/deal/:id)

Shows a single deal's full copilot intelligence view with health score ring, DHI flags, compliance status, approval timeline with actor/timestamp/rationale audit trail, and configuration details.

### 7.4 AE Pipeline (/pipeline)

Kanban-style board with columns: Draft → Pre-Check → In Review → Approved / Rejected. Each card shows deal name, customer, value, health score, flag count.

### 7.5 Customer Dashboard (/customer)

Shows "My Proposals" with cards for each submitted proposal. Each card shows proposal name, status badge (with color), submission date, value, and health score. Auto-approved proposals show a "⚡ AUTO-APPROVED" badge.

## 7.6 Customer Submit Proposal (/customer/submit)

Similar to Quote Builder but from customer perspective. Includes the same 8 validation rules firing in real-time. Shows a prominent Auto-Approval Eligibility Banner when the proposal qualifies for instant approval (score  $\geq 90$ , no critical flags, value under threshold). Uses the same Nexlara SKU catalogue (66 items).

## 7.7 Customer Proposal Detail (/customer/proposal/:id)

Shows proposal status, approval timeline with actor/timestamp/rationale audit trail, DoA tier info, Oracle CPQ reference IDs, product configuration table, compliance panel, and DHI flags status.

## 7.8 Customer History (/customer/history)

Summary metrics (total proposals, approval rate, avg cycle time) + historical proposals table with filtering.

# 8. Data Model

TypeScript

```
interface Deal {
  id: string;           // Quote ID (e.g., "Q-2025-00079")
  name: string;         // Deal name (e.g., "Montreal Services – Analytics Net
  customer: string;     // Customer name
  value: number;        // Total ACV in EUR
  currency: string;     // "EUR"
  healthScore: number;  // 0-100 DHI score
  stage: "draft" | "pre-check" | "in-review" | "approved" | "rejected";
  daysInStage: number;
  ae: string;           // Account Executive name
  region: string;       // Geographic region (EMEA-DACH, NA-East, etc.)
  dealMotion: string;   // "Net New" | "Upsell" | "Cross-sell" | "Renewal" | "R
  licenseModel: string;
  contractMonths: number;
  discountPercent: number;
  bomLines: number;
  dhiFlags: DHIFlag[];
  approvalProbability: number; // ML score 0-100
  resubmissionCount: number;
  doaTier: string;
  lastUpdated: string;
  timeline: TimelineEvent[];
}
```

```

interface DHIFlag {
  code: string;           // e.g., "DHI-C-005"
  severity: "critical" | "warning" | "info";
  title: string;
  description: string;
  resolved: boolean;
}

interface TimelineEvent {
  action: string;
  actor: string;
  timestamp: string;
  detail: string;
}

interface SKU {
  sku: string;             // e.g., "NXL-ERP-ENT-001"
  name: string;
  family: string;         // "ERP" | "HCM" | "Analytics" | "AI Add-on" | "Support
  lepPrice: number;
  licenseModel: string;
  status: string;         // "Active" | "Deprecated"
  successorSku?: string;
}

```

## Sample Data

The app uses 6 representative deals from the sample data package spanning all stages and geographies, plus a 66-item Nexlara SKU catalogue covering ERP, HCM, Analytics, AI Add-on, and Support product families.

## 9. Key Behavioral Requirements

1. Every state change triggers recalculation — adding a SKU, moving a slider, typing in justification all cause the intelligence panel to update instantly
2. Rule Activity Bar flashes for 2 seconds showing which rule just fired
3. "Resolve Inline" buttons fix issues without leaving the page or opening external ticketing systems
4. Score factors show WHY — not just a number, but "2 critical DHI flags (-50)" and "Strong justification (+5)"
5. Auto-approval path is clearly distinct from human-reviewed path

6. Toast notifications confirm every inline resolution action
  7. Oracle CPQ references appear throughout (system of record messaging, quote IDs, opportunity IDs)
  8. DoA framework is non-negotiable — always shown, always respected above thresholds
  9. DHI is a non-negotiable BLOCKER — no quote proceeds with unresolved critical flags
  10. AI Agents Processing modal shows the full 2-layer orchestration on submission
- 

## 10. Hard Constraints

1. **Oracle CPQ remains system of record** — the UI is an extension layer, not a replacement
  2. **Delegation of Authority (DoA) framework is non-negotiable** above defined thresholds
  3. **Full auditability** — every approval, bypass, waiver, DHI resolution logged with actor + timestamp + rationale
  4. **ASC 606 compliance is non-negotiable** — must validate at configuration time
  5. **DHI is a non-negotiable BLOCKER** — unresolved critical/high flags block submission regardless of deal value
  6. **Seller experience cannot worsen** — the copilot reduces effort, doesn't add steps
  7. **License model transition** — must handle perpetual, subscription, consumption, and hybrid
  8. **MCCA must be signed** — no quote submission without active Master Cloud Customer Agreement
  9. **Deterministic-first architecture** — Layer 1 rules make all binding decisions, LLM is advisory only
- 

## 11. File Structure

Plain Text

```
client/  
  index.html          ← Google Fonts (Space Grotesk, Inter, JetBrains Mono)  
  src/  
    App.tsx           ← Routing (AERouter + CustomerRouter based on path pr  
    index.css         ← Tailwind 4 + OKLCH dark theme variables + custom ut  
    lib/  
      data.ts         ← TypeScript interfaces, 6 sample deals, 66-item SKU  
    components/
```

```
Layout.tsx          ← AE sidebar navigation
CustomerLayout.tsx  ← Customer sidebar navigation
AIAgentsProcessing.tsx ← 2-layer orchestrator modal (Rules Engine → Skill
ui/                  ← shadcn/ui components (pre-installed)
pages/
  Dashboard.tsx      ← AE command center
  QuoteBuilder.tsx   ← PRIMARY: AE quote config with 8 rules
  DealDetail.tsx     ← AE deal detail view
  Pipeline.tsx       ← AE pipeline kanban
  Compliance.tsx     ← AE compliance view
  Analytics.tsx      ← AE analytics
customer/
  CustomerDashboard.tsx ← Customer proposals list
  SubmitProposal.tsx   ← Customer proposal form with validation
  ProposalDetail.tsx   ← Customer proposal status/audit trail
  History.tsx          ← Customer history view
```

## 12. Dependencies (package.json)

Key dependencies: react, react-dom, wouter, tailwindcss, framer-motion, recharts, lucide-react, sonner, class-variance-authority, clsx, tailwind-merge, @radix-ui/\* (via shadcn/ui), zod

## 13. Logo & Branding

- **Logo:** Geometric "N" mark in Electric Violet (#A100FF) on transparent background
- **Brand name:** NEXLARA (uppercase, Space Grotesk Bold, tracked tight)
- **Tagline for AE module:** "Deal Health Command Center"
- **Tagline for Customer module:** "Customer Portal"
- **Platform badge:** "Oracle CPQ" (shown in header)
- **System indicator:** "● 8 RULES ACTIVE" (shown in header)

## 14. Change Log (v1.0 → v2.0)

| Area               | Previous (v1.0)         | Updated (v2.0)                |
|--------------------|-------------------------|-------------------------------|
| Platform           | D365 CPQ / Dynamics 365 | Oracle ERP Cloud + Oracle CPQ |
| Validation engines | 6 engines               | 8 deterministic rules         |

|                   |                              |                                                                        |
|-------------------|------------------------------|------------------------------------------------------------------------|
| Architecture      | Flat 6-engine model          | 2-layer: Rules Engine + Skills Layer                                   |
| DHI               | One of 6 engines             | Dedicated BLOCKER rule (non-negotiable)                                |
| MCCA              | Not included                 | Added as BLOCKER rule                                                  |
| Auto-Attach       | Part of DHI                  | Separate WARNING rule                                                  |
| Deprecated SKU    | Part of DHI                  | Separate WARNING rule                                                  |
| DoA Tiers         | Tier 0–4 (generic)           | AUTO_APPROVE / FIRST_APPROVER / SUPER_APPROVER / MU_CFO / BOARD        |
| Integration       | Dataverse Plugin SDK + OData | Oracle BML + REST API + OIC                                            |
| ServiceNow        | Used for ticket routing      | Eliminated entirely                                                    |
| Product catalogue | 14 Microsoft SKUs            | 66 Nexlara SKUs (ERP, HCM, Analytics, AI, Support)                     |
| Submission UX     | Direct submit                | AI Agents Processing modal showing orchestrator execution              |
| Skills Layer      | Not present                  | LLM-powered advisory (DHI Translator, Narrative Advisor, ML Explainer) |
| Audit             | Basic logging                | Immutable audit with orchestrator result schema                        |